

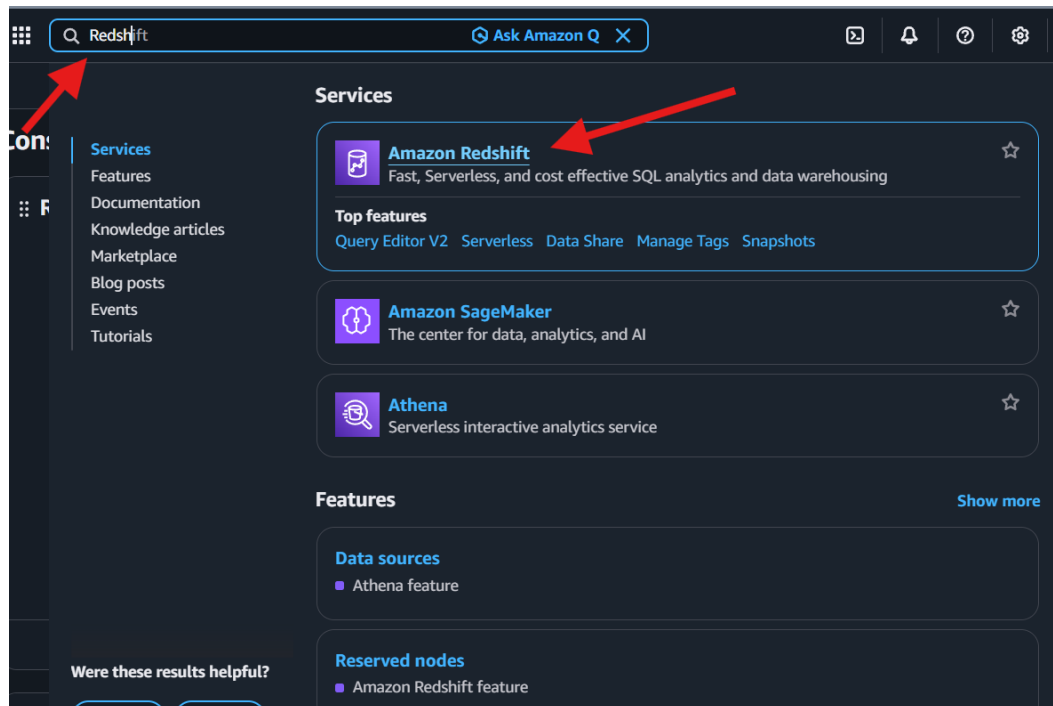
Creating an Amazon Redshift Provisioned Cluster

To Begin with the Lab

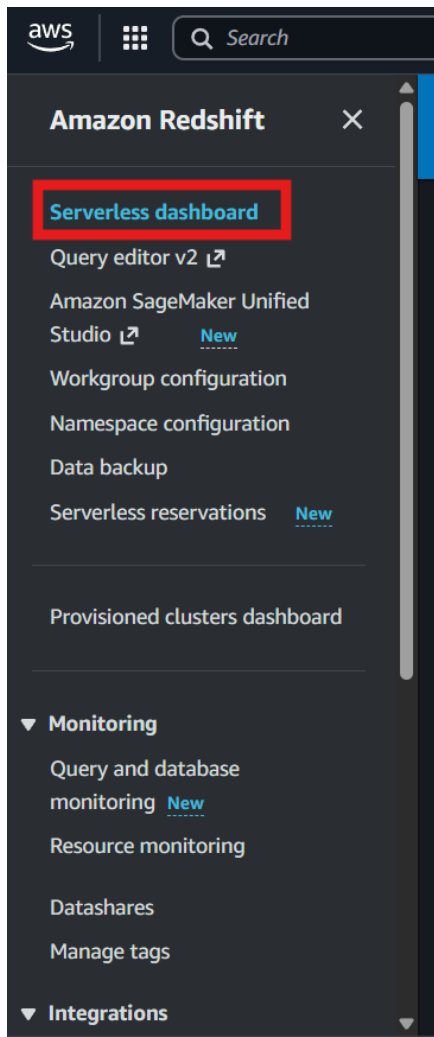
Summary of the Lab

In this lab, a provisioned cluster was created in **Amazon Redshift** for running analytical workloads and sample queries. The process included navigating from the Redshift Serverless dashboard to the Provisioned Clusters dashboard, configuring a cost-effective RA3 node type, loading the Tackit sample dataset, and applying the default cluster permissions and networking configuration. Default database settings were used, and an administrator password was generated automatically. After the cluster creation request was submitted, the cluster status was monitored until it became available. Finally, the cluster details page was opened to verify the successful deployment and review the cluster configuration.

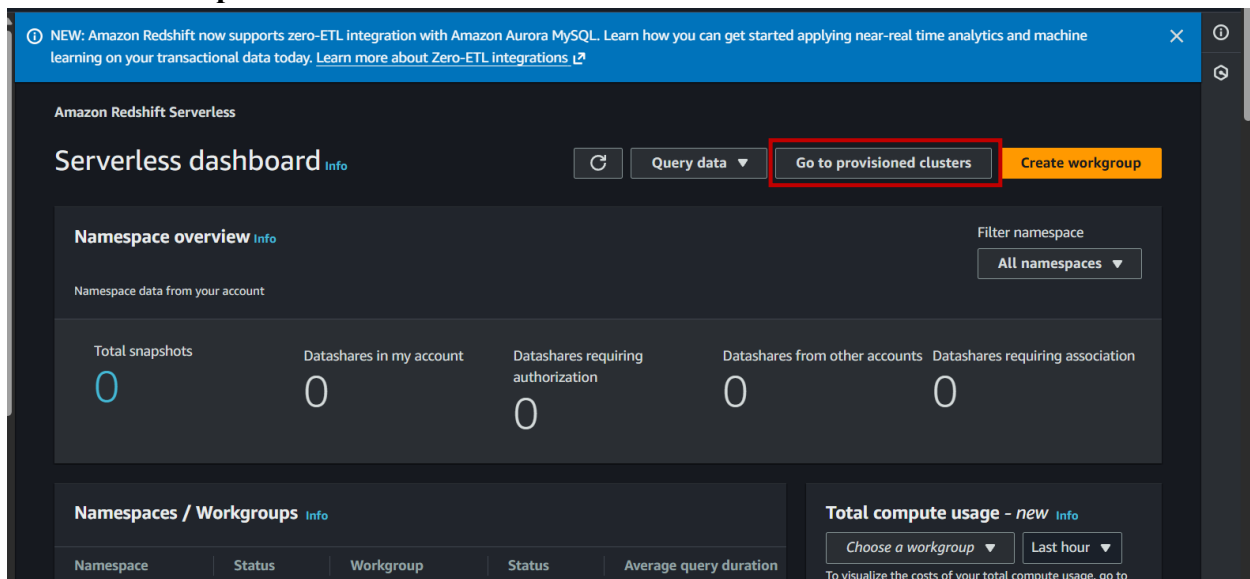
- Sign in to the AWS Management Console
- Search for **Amazon Redshift** in the AWS search bar



- Open **Amazon Redshift**
- Navigate to **Serverless dashboard**



- Click **Go to provisioned clusters**



- Navigate to the **Provisioned Clusters Dashboard**

- Click **Create cluster**

Provisioned clusters dashboard

Try Amazon SageMaker Unified Studio [↗](#) Go to Serverless [↗](#) Purchase reserved nodes **Create cluster**

Resources overview
Resource data for Asia Pacific (Mumbai) Region.

Total nodes	On-demand nodes	Reserved nodes	Reserved nodes available (0 of 0 used)	Automated snapshots	Manual snapshots
0	0	0	0	0	0

Cluster overview (0) Any Status ▼

Cluster	Status	Average query duration ▼	Average number of queries run daily ▼
No clusters			

Create an Amazon Redshift cluster

Datashares
Authorize other AWS accounts to access datashares created in this AWS account. Associate or decline datashares from other AWS accounts.

[Require authorization](#)

0

- Keep the default cluster identifier
- Select **ra3.large** as the node type

☐ Help me choose

Node type [Info](#)

Choose a node type that meets your CPU, RAM, storage capacity, and drive type requirements.

ra3.large

rg.xlarge
Managed storage: up to 32 TB/node
\$0.865/node/hour \$0.026/GB/month

rg.4xlarge
Managed storage: up to 128 TB/node
\$3.459/node/hour \$0.026/GB/month

RA3
High performance with scalable managed storage

ra3.large ✓
Managed storage: up to 8 TB/node
\$0.617/node/hour \$0.026/GB/month

ra3.xlplus
Managed storage: up to 4 TB for single-node clusters. For clusters with 2 or more nodes, each node has storage up to 32 TB.
\$1.235/node/hour \$0.026/GB/month

\$450.41/month **1 TB** **\$0.026/GB/month**

- Keep the default cluster configuration settings
- Set the number of nodes to **1**

Number of nodes
Enter the number of nodes that you need.

1

Range (1-16)

Configuration summary [Info](#)
ra3.large | 1 node

\$450.41/month Estimated on-demand compute price Save more than 60% of your costs by purchasing reserved nodes. Learn more about pricing ↗	1 TB Max compressed storage RA3 stores data in Redshift managed storage. Each RA3.large node gets up to 8 TB of compressed data capacity in managed storage to ensure optimal query performance.	\$0.026/GB/month Estimated storage price Pay only for the amount of data you store in managed storage when running an RA3 cluster.
--	--	--

- Understand that a single node cluster acts as both the leader node and compute node
- Enable **Load sample data**
- Verify that the Tickit sample dataset is selected automatically

Sample data [Info](#)

☒ **Load sample data**
Load sample data to your Redshift cluster to start using the query editor to query data.

Tickit (28 MB)

Tickit is the sample data set that uses a sample database called TICKIT. Tickit contains individual sample data files: two fact tables and five dimensions.

Database configurations

Admin user name
Enter a login ID for the admin user of your DB instance.

awsuser

The name must be 1-128 alphanumeric characters, and it can't be a [reserved word](#) ↗.

Admin password
Select an option to manage your admin password.

- Keep the default database configuration settings
- Click **Generate password**

Database configurations

Admin user name
Enter a login ID for the admin user of your DB instance.

awsuser

The name must be 1-128 alphanumeric characters, and it can't be a [reserved word](#).

Admin password
Select an option to manage your admin password.

☐ Manage admin credentials in AWS Secrets Manager [Info](#)
AWS manages a KMS key that encrypts your data.

☒ **Generate a password**
Amazon Redshift generates an admin password.
Admin user password
[password field]
☐ Show password

☐ Manually add the admin password
Manually enter the admin password.

Choose a key type
Encrypt all the data on your cluster. Choose one of the following:

☒ **AWS owned key**
A KMS key that AWS services creates in Redshift service account.

- Copy and save the generated administrator password
- Scroll to the **Cluster permissions** section
- Select the default IAM role and permissions configuration

Cluster permissions

[i](#) Create an IAM role as the default for this cluster that has the [AmazonRedshiftAllCommandsFullAccess](#) policy attached. This policy includes permissions to run SQL commands to COPY, UNLOAD, and query data with Amazon Redshift. The policy also grants permissions to run SELECT statements for related services, such as Amazon S3, Amazon CloudWatch logs, Amazon SageMaker, and AWS Glue.

Associated IAM roles (1) [Info](#)

[Set default](#) [Actions](#) [Associate IAM roles](#) [Create IAM role](#)

Create, associate, or remove an IAM role. You can associate up to 50 IAM roles. You can also choose an IAM role and set it as the default for this cluster.

< 1 > [Settings](#)

☐	IAM roles	Status	Role ty
☐	AmazonRedshift-CommandsAccessRole-20260523T14...	Not applied	Default

- Navigate to the **Register with AWS Glue Data Catalog** section
- Select **Do not register with Amazon Redshift federated permissions**
- Keep the **Use Defaults** on to prevent the use of additional configurations

Register with AWS Glue Data Catalog
Enable unified permissions across data warehouse and data lake environments by registering your Amazon Redshift resource with AWS Glue Data Catalog. [Learn more](#)

☐ Register with Amazon Redshift federated permissions - *new*
Enable fine-grained access control with Amazon Redshift policies like Column-Level privileges (CLP), Row-Level security (RLS), and Dynamic Data Masking (DDM).

☒ Do not register with Amazon Redshift federated permissions
Select registration options later.

Additional configurations ☒ Use defaults

These configurations are optional, and default settings have been defined to help you get started with your cluster. Turn off "Use defaults" to modify these settings now.

i The default configuration has been updated to **default.redshift-2.0**. This configuration enforces SSL connections for enhanced security.

Network Backup

- Keep the default networking, VPC, security group, and backup settings
- Scroll to the bottom of the configuration page
- Click **Create cluster**

Additional configurations ☒ Use defaults

These configurations are optional, and default settings have been defined to help you get started with your cluster. Turn off "Use defaults" to modify these settings now.

i The default configuration has been updated to **default.redshift-2.0**. This configuration enforces SSL connections for enhanced security.

Network
Using default VPC (vpc-072c4acd761c3b942) and default subnet.

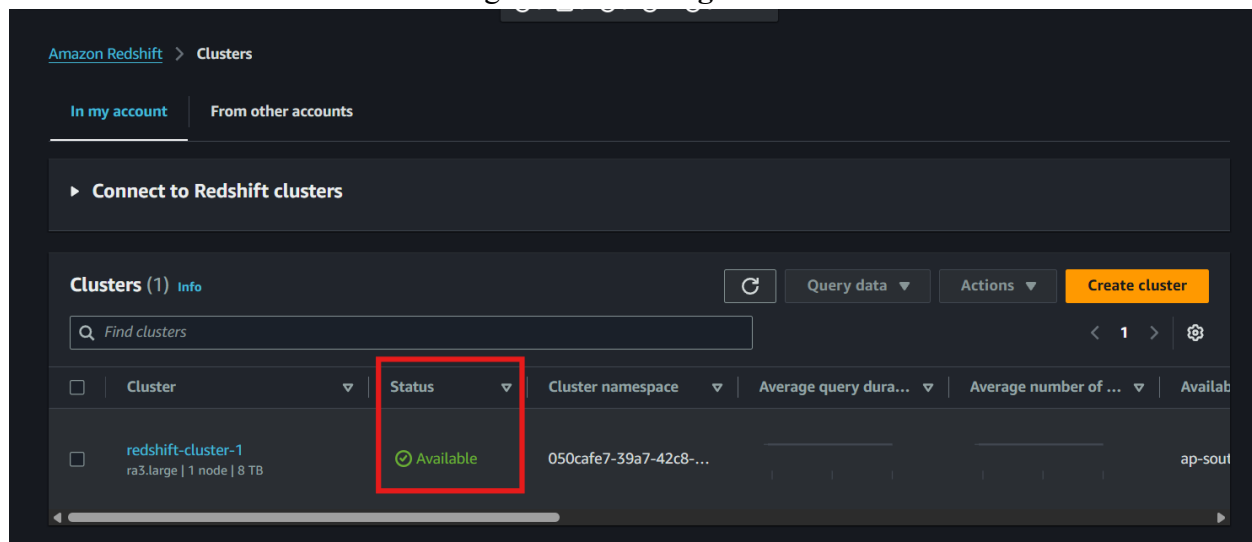
Backup
Automated snapshots are created about every eight hours or following every 5 GB per node of data changes, whichever comes first.

Security
Using default (sg-05a49708dcf9142a5) cluster security group.

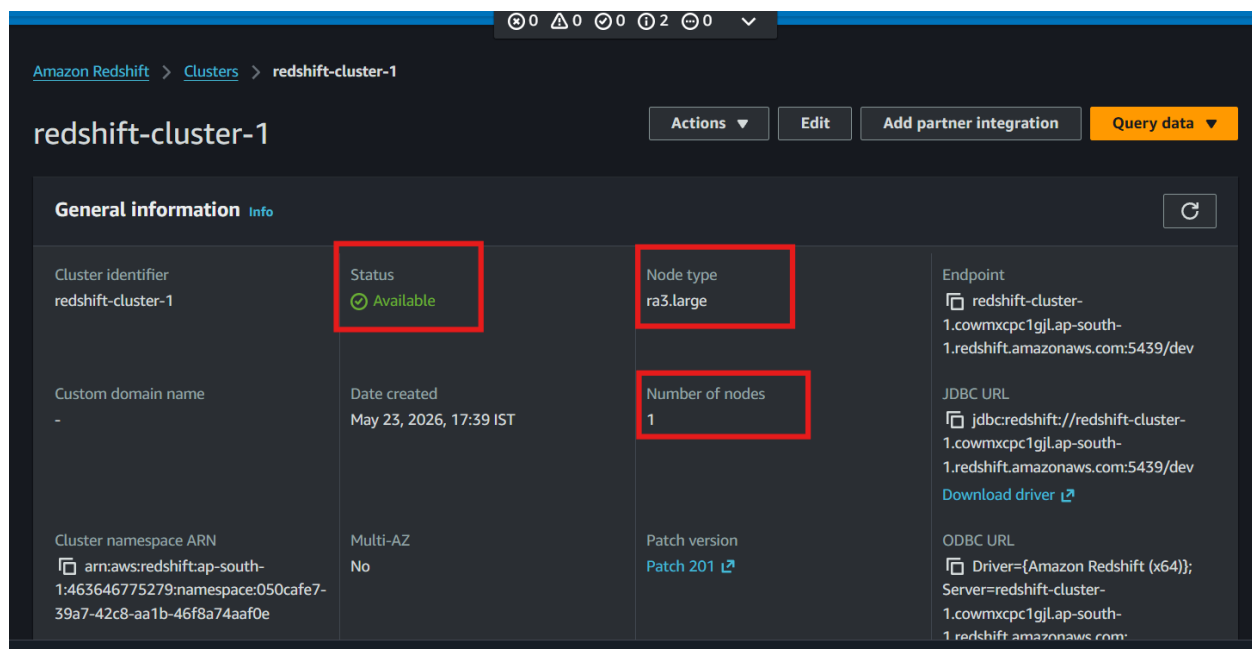
Configuration
Using default.redshift-2.0 parameter group with no database encryption.

Cancel **Create cluster**

- Wait for the cluster status to change from **Creating** to **Available**



- Click the cluster name after deployment completes
- Review the cluster details including cluster identifier, status, node type, endpoint, and number of nodes



- Verify that the cluster status is **Available** and the provisioned cluster has been created successfully